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Customer Churn Classification in Telecom Sector

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ABSTRACT

The telecommunications sector faces fierce competition, making customer retention vital. The loss of subscribers, or churn, results in significant revenue decline. This project aims to leverage data mining techniques to build predictive models for Customer Churn Classification in the telecom industry.

We utilize a comprehensive dataset of customer demographics, usage history, and contract information to identify the key features driving churn behavior. The study employs and compares several Machine Learning algorithms, specifically Logistic Regression, Decision Trees,Gradient Boosting,Random Forest and K-Nearest Neighbours(KNN). The models are evaluated using metrics like Accuracy, F1-Score, and AUC-ROC to determine the optimal classifier.

1.0 Introduction

1.1 Background and Motivation

The global telecommunications industry is one of the most dynamic and competitive sectors in the modern economy. With deregulation and technological advancements, customers face minimal barriers to switching service providers, resulting in a phenomenon known as customer churn. This voluntary departure of subscribers represents a critical challenge for telecom companies, leading directly to substantial revenue loss, increased costs for customer acquisition, and erosion of market share. Studies suggest that acquiring a new customer can cost significantly more than retaining an existing one. Therefore, the ability to predict and prevent churn has become a paramount strategic objective for sustaining profitability and long-term viability in this saturated market.

1.2 Statement of Problem

Customer churn is a complex, multi-faceted problem driven by a variety of factors, including service quality, pricing, competitor offers, and customer satisfaction. The challenge lies in identifying at-risk customers *before* they make the final decision to leave. Without an automated, data-driven system, identifying these subtle indicators across millions of customer records is virtually impossible, leading to reactive and often ineffective retention efforts.

1.3 Objective

1.3.1 General Objective

The primary objective is to develop a highly accurate and robust classification model capable of predicting which customers are most likely to churn.

1.3.2 Specific Objective

- **Pre-process and explore**: a real-world telecom customer dataset to uncover hidden patterns and identify key predictive features associated with churn.
- **Implement and evaluate :** various state-of-the-art Machine Learning algorithms, including Logistic Regression, Decision Trees, and K-Nearest Neighbours(KNN), Random Forest and Gradient Boosting for the binary classification task.
- **Determine the optimal model :** based on rigorous performance evaluation using industry-standard metrics such as Accuracy, Precision, Recall, F1-Score, and AUC-ROC.
- **Provide actionable insights** and feature importance analysis that can guide the development of targeted, cost-effective retention campaigns

2.0 Literature Review

Customer churn prediction is a vital aspect of customer relationship management (CRM) in the telecom industry. The ability to accurately predict which customers are likely to leave allows companies to implement targeted retention strategies, thereby minimizing revenue losses. Over the years, various methods and techniques have been developed and applied to predict churn, ranging from statistical models to advanced machine learning algorithms. This literature survey reviews key studies and methodologies in the domain of customer churn prediction, focusing on the application of machine learning techniques. The overall problem is a complex challenge in the telecom industry requiring comprehensive analysis.

2.1 Traditional Methods and Early Models

Early studies on churn prediction primarily relied on statistical methods and traditional data mining techniques. Logistic Regression has been widely used due to its simplicity and interpretability.

- Tsai and Lu (2009) demonstrated the use of Logistic Regression in predicting customer churn, highlighting its effectiveness in identifying key predictors and providing interpretable results.
- Similarly, Burez and Van den Poel (2009) explored class imbalance in churn prediction, showcasing how balancing techniques can improve model performance.

2.2 Decision Trees and Ensemble Methods

With the advancement in computing power and the availability of large datasets, more complex models like Decision Trees and ensemble methods gained popularity.

- Breiman (2001) introduced Random Forests (RF), an ensemble method that constructs multiple decision trees during training and outputs the mode of the classes for classification.
- This method improves predictive performance and robustness against overfitting.
- Liaw and Wiener (2002) further demonstrated the application of Random Forests in various domains, including churn prediction, emphasizing its accuracy and stability.
- Idris, Rizwan, and Khan (2012) applied Random Forests to predict churn in the telecom industry, highlighting its superior performance compared to single-tree methods.
- The use of ensemble methods like Random Forests and Gradient Boosting has become a standard approach in churn prediction due to their ability to handle large, complex datasets and improve prediction accuracy.

2.3 Advanced Machine Learning Techniques

Gradient Boosting, introduced by Friedman (2001), is another powerful ensemble method that builds models sequentially, with each new model correcting errors made by previous ones.

- Chen and Guestrin (2016) developed XGBoost, a scalable implementation of Gradient Boosting that has been widely adopted for its efficiency and performance in large-scale datasets.
- Recent studies have demonstrated the effectiveness of Gradient Boosting in churn prediction.
- Sharma and Panigrahi (2013) applied Gradient Boosting to predict customer churn in the telecom industry, achieving high accuracy and demonstrating its superiority over traditional methods.
- Mishra and Reddy (2017) compared various ensemble classifiers, including Gradient Boosting, for churn prediction, highlighting its robustness and superior predictive power.

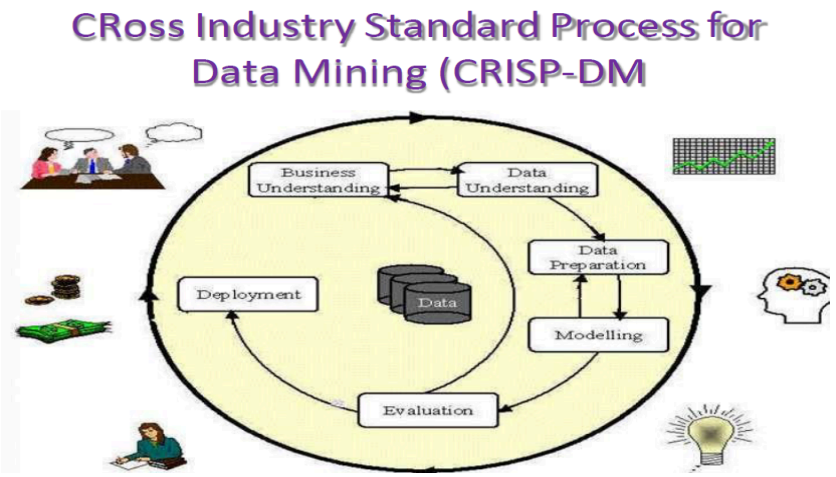
- Further research focuses on the application of data mining techniques in the telecom industry to solve customer churn prediction problems.

2.4 Hybrid and Deep Learning Models

The integration of multiple models, known as hybrid models, has also been explored in churn prediction.

- These models combine the strengths of different algorithms to improve predictive performance.
- Huang and Kachadi (2013) developed a hybrid learning system for telecom churn prediction, combining decision trees and neural networks to achieve better accuracy.
- Deep learning, particularly neural networks, has gained attention for its ability to model complex, non-linear relationships in data.
- Lu et al. (2014) implemented a hybrid model combining neural networks with traditional classifiers for churn prediction, achieving significant improvements in accuracy.

3.0 Approach and methodology



The project's methodology is structured around the Cross-Industry Standard Process for Data Mining (CRISP-DM), which organizes the data mining process into six iterative phases.

1. Business Understanding

Focuses on understanding the **project objectives** and requirements from a business perspective. It translates business problems (like high churn) into data mining goals.

2. Data Understanding

Involves collecting the initial data, immersing in it, and identifying data quality issues, interesting subsets, and initial insights or hypotheses.

3. Data Preparation

The crucial step of cleaning, transforming, and formatting the data into the final dataset ready for modeling. This includes handling missing values, encoding, and scaling features.

4. Modeling

Involves selecting and applying various data mining techniques (like Logistic Regression, Random Forest, etc.), and calibrating their parameters to create the models.

5. Evaluation

Assesses the model(s) rigorously and determines if the results meet the initial business objectives. This phase validates the solution before deployment.

6. Deployment

The final phase where the developed model is put into practical use to generate predictions (e.g., scoring new customers weekly) and the results are integrated into the business process.

Business Understanding

This section illustrates the initial process steps of the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework as applied to the Customer Churn Classification project. The first and most critical phase is Business Understanding.

1. Business Understanding

The CRISP-DM process begins with a clear definition of the **business problem**. In the telecom sector, the core business problem is the financial and operational impact of customer attrition. Our business problem can be articulated through the following key questions:

1. What is the impact of customer churn on the company's revenue and market share? (The core motivation)
2. What are the key customer features and behaviors that most strongly predict a customer will churn? (Identifying predictive factors)
3. Can we accurately classify high-risk customers early enough to implement targeted, cost-effective retention strategies? (The objective of the machine learning solution)

2.Data Understanding

We used data from the kaggle and our attributes are 20

Attribute	Missing Value	Description
1.customerID	0	Unique customer identifier
2.Gender	0	Male or Female
3. `SeniorCitizen`	0	1 if the customer is a senior citizen, 0 otherwise
4. `Partner`	0	Whether the customer has a partner (Yes/No)
5. `Dependents`	0	Whether the customer has dependents (Yes/No
6. `tenure`	0	Number of months the customer has stayed with the company

7. PhoneService	0	Whether the customer has phone service (Yes/No)
8. MultipleLines	0	Whether the customer has multiple lines (No, Yes, No phone service)
9. InternetService	0	Internet service provider (DSL, Fiber optic, No)
10. OnlineSecurity	11	Whether the customer has online security (Yes, No, No internet service)
11.Online Backup	0	Whether online backup service is included
12.DeviceProtection	0	Whether device protection is included
13.TechSupport	0	Whether technical support is included

14.StreamingTV	0	Whether streaming TV service is included
15.StreamingMovies	0	Whether streaming movie service is included
16.Contract	0	Type of Contract(Month to month,One Year,two year
17.PaperlessBilling	0	Whether billing is paperless
18.PaymentMethod	0	Method of payment (Credit card, Bank transfer, Mailed check, etc.)
19.MonthlyCharges	11	Amount charged per month
20.TotalCharges	11	Total amount charged over the customer’s lifetime

The 21th attribute(churn) represents the **class label** (what we want to predict)

Measurement Scales for our data

Nominal scale (categories, no order):
customerID, Gender, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, PaperlessBilling, PaymentMethod, **Churn**

Ordinal scale (ordered categories, no fixed distance):
Contract (Month-to-month < One year < Two year)

Interval / Numeric scale (real numerical values):
SeniorCitizen, tenure, MonthlyCharges, TotalCharges(after empty spaces are filled with 0.0)

3.2.2: Data Quality and Cleaning

Data quality assessment is a critical phase to ensure the reliability and validity of the machine learning models. During the preliminary data exploration of the filtered 20-attribute dataset, two primary data quality issues were identified and addressed: missing values and incorrect data types.

1.Handling Missing Values in TotalCharges column.

As seen during the analysis, we encountered **11 missing values** in the TotalCharges column.

- **Diagnosis:** Upon cross-referencing these missing entries with other attributes, it was discovered that all 11 records corresponded to customers with a tenure of **0 months**. This is a logical issue, as customers who have just joined (0 months) have not yet accrued any TotalCharges and the data entry was left blank.
- **Resolution:** To ensure data integrity and avoid discarding valid customer records, the missing values in TotalCharges were imputed by filling the NaN (Not a Number) entries with **0.0**. This method is appropriate because zero tenure logically dictates zero cumulative charges.

2. Handling Missing Values in MonthlyCharges

Diagnosis: To simulate a real-world data collection error and test imputation techniques, 11 values were intentionally deleted from the MonthlyCharges column, creating 11 missing entries (NaN). This attribute is a **continuous numerical feature**.

Resolution: The missing values in MonthlyCharges were imputed by filling the NaNentries with the **median** of the column.

- **Imputation Technique: Median Imputation**
- **Justification:** The median is the preferred method for imputing missing values in numerical columns like MonthlyCharges because it is **robust to outliers**. Unlike the mean (average), the median is not heavily influenced by extremely high or low monthly charges, thus minimizing the distortion introduced to the column's distribution.

3. Handling Missing Values in OnlineSecurity

Diagnosis: To further test categorical imputation techniques, 11 values were intentionally deleted from the OnlineSecurity column, creating 11 missing entries (NaN). This attribute is a **nominal categorical feature** (containing values like "Yes," "No," or "No internet service").

Resolution: The missing values in OnlineSecurity were imputed by filling the NaN entries with the **mode** (most frequent category) of the column.

- **Imputation Technique: Mode Imputation**
- **Justification:** For categorical variables, the mode is the statistically appropriate measure of central tendency. Filling the missing data with the most frequently occurring category (e.g., "No internet service" or "No") is a simple and effective technique that preserves the overall distribution of the categorical feature.

4. <u>**Feature Exclusion (Dropping Attributes)**</u>

- The **`customerID`** column was dropped from the dataset. This attribute is a unique administrative identifier with no predictive value and would only add noise and complexity to the machine learning model.

Data Transformation

1. Objective

To prepare the cleaned Telco Churn dataset for machine learning by converting categorical text attributes into numerical formats and ensuring all mathematical features are correctly typed.

2. Transformation Steps

A. Numerical Enforcement

Before encoding, continuous variables were explicitly cast to numeric types.

- Attributes: TotalCharges, MonthlyCharges.
- Action: Used pd.to_numeric with errors='coerce' to handle potential formatting errors, followed by .fillna(0) to ensure no null values remained. This prevented the creation of thousands of unnecessary columns.

B. One-Hot Encoding (OHE)

To handle nominal categorical data (where no natural order exists), One-Hot Encoding was applied.

- Method: pd.get_dummies().
- Logic: For every unique value in a category (e.g., "Contract"), a new binary column is created.
- Optimization: drop_first=True was used to avoid the Dummy Variable Trap (multicollinearity). If a feature has N categories, it is represented by N-1 columns.

3. Data Shape Evolution

The transition of the data structure followed this logic:

Stage	Column Count	Explanation
Pre-Transformation	20	Cleaned data after dropping customerID.

Post-Transformation	~31	Result of expanding categories like InternetService and Contract into multiple binary columns.

4. Final Transformed Attributes (Sample)

The final dataset (df_final) now consists entirely of:

- Integers/Floats: tenure, MonthlyCharges, TotalCharges.
- Booleans/Dummies: gender_Male, Contract_One year, Contract_Two year, InternetService_Fiber optic.

1. What is Classification?

Classification is a supervised machine learning task where the goal is to assign data points to predefined categories (classes) based on input features.

In simple terms:

- The model learns from labeled data
- Then predicts the class for new, unseen data

Example in Telecom Churn:

- Input features: tenure, monthly charges, contract type, internet service, etc.
- Target (class label):
    - 0 → Customer Stayed
    - 1 → Customer Churned

So, churn prediction is a binary classification problem because there are only two possible outcomes.

2. Why We Used CRISP-DM Methodology

We used CRISP-DM (Cross-Industry Standard Process for Data Mining) because it provides a structured, systematic, and industry-accepted framework for data science projects.

CRISP-DM Phases and Their Role in This Project:

1. Business Understanding
    - Goal: Reduce customer churn in the telecom sector
    - Importance: Churn directly affects revenue and customer lifetime value
2. Data Understanding
    - Explored the telecom customer dataset
    - Identified missing values, categorical features, and churn imbalance
3. Data Preparation
    - Cleaned data (handled missing values)
    - Encoded categorical variables
    - Scaled numerical features
    - Prepared data for machine learning models
4. Modeling
    - Applied multiple classification algorithms
    - Tuned hyperparameters
    - Compared model performance
5. Evaluation
    - Used accuracy, precision, recall, F1-score

    - Visualized results using confusion matrices

6. Deployment (Conceptual)
    - Best-performing model can be deployed to predict churn in real-time

3. Why We Selected 5 Algorithms for Churn Classification

Using multiple algorithms helps:

- Compare performance
- Reduce model bias
- Select the most reliable model for business decisions

The 5 Classification Algorithms Used

1. Logistic Regression

- Baseline classification model
- Easy to interpret
- Helps understand feature impact on churn

2. Decision Tree

- Rule-based and easy to visualize
- Captures non-linear relationships
- Useful for business explainability

3. Random Forest

- Ensemble of multiple decision trees

- Reduces overfitting
- High accuracy and robustness

4. K-Nearest Neighbors (KNN)

- Distance-based classifier
- Works well after feature scaling
- Useful for pattern similarity detection

5. Gradient Boosting

- Sequential ensemble model
- Focuses on correcting previous errors
- Often delivers the best predictive performance

Why These 5?

1. Mix of simple and advanced models
2. Covers linear, tree-based, distance-based, and ensemble methods
3. Enables strong comparison and validation

4. Tools and Technologies Used

Programming Language

- Python
    - Simple syntax
    - Extensive machine learning ecosystem
    - Industry standard for data science

Development Environments

- **Anaconda**
    - Manages Python environments and packages
- **Google Colab**
    - Cloud-based execution
    - No local setup required
- **Visual Studio Code (VS Code)**
    - Professional IDE
    - Code debugging and version control support

Python Libraries Used

1. Pandas

- Data loading and manipulation
- Handling missing values
- Data cleaning and preprocessing

2. NumPy

- Numerical computations
- Efficient array operations
- Supports mathematical functions

3. Matplotlib

- Data visualization

- Confusion matrices
- Model performance plots

4. Scikit-learn (sklearn)

- Machine learning algorithms
- Train-test splitting
- Model evaluation metrics
- Classification models:
    - Logistic Regression
    - Decision Tree
    - Random Forest
    - KNN
    - Gradient Boosting

Meaning of Each Algorithm (What it is)

1. Logistic Regression

A statistical classification algorithm that models the probability of a binary outcome using a sigmoid function.
It estimates how each feature increases or decreases the probability of churn.

2. Decision Tree

A tree-based model that splits data using a sequence of if–else rules.
The model learns decision rules such as:

"If contract is month-to-month AND tenure is low → churn"

3. Random Forest

An ensemble of many decision trees where each tree votes for a class.
Reduces overfitting and improves stability by averaging multiple trees.

4. Gradient Boosting

An ensemble model that builds trees sequentially, where each new tree corrects the errors of the previous ones.

Focuses more on difficult-to-predict customers, especially churners.

5. K-Nearest Neighbors (KNN)

A distance-based algorithm that classifies a customer based on the majority class of its nearest neighbors.

Customers with similar characteristics are assumed to behave similarly.

What Each Algorithm Does in THIS Project

Project Goal

To **predict customer churn (Yes / No)** for a telecom company using customer demographic, service, and billing data.

Logistic Regression (Best overall)

```
Logistic Regression Accuracy: 80.48%

Confusion Matrix:
[[925 110]
 [165 209]]

Classification Report:
              precision    recall  f1-score   support

       False       0.85      0.89      0.87      1035
        True       0.66      0.56      0.60       374

    accuracy                           0.80      1409
   macro avg       0.75      0.73      0.74      1409
weighted avg       0.80      0.80      0.80      1409

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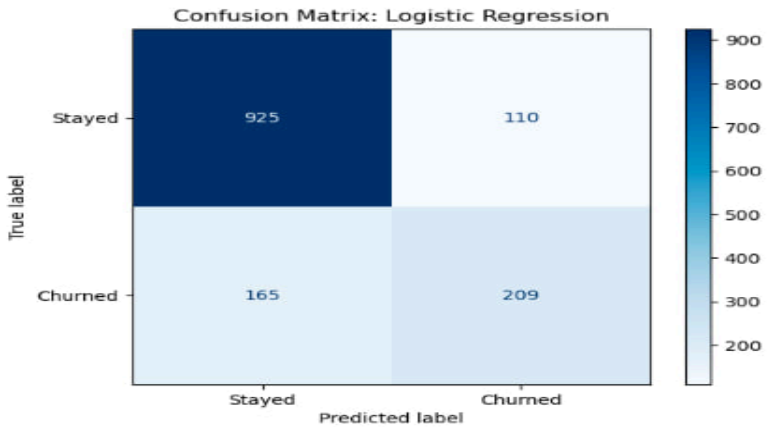


- Treats churn as a **probability problem**
- Learns linear relationships between features and churn

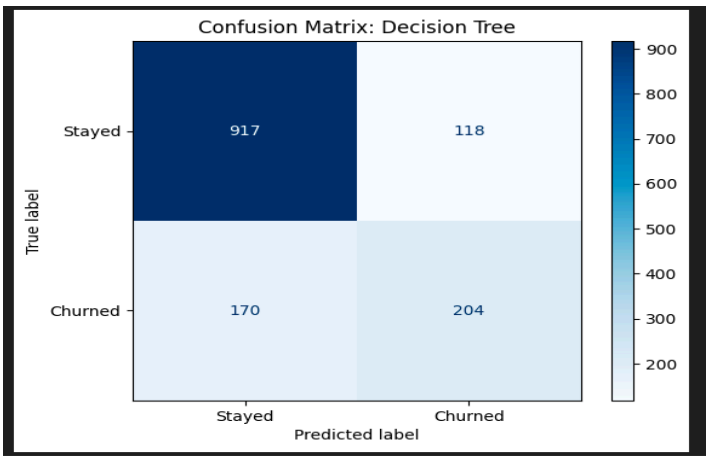
- Performs well when data is well encoded and scaled

Results:

- **Accuracy:** 80.48%
- **Churn Recall:** 56% (best among all)
- **Churn F1-score:** 0.60 (highest)

Best at identifying customers who will leave

Decision Tree



- Uses clear rule-based splits
- Easy to interpret and explain to business stakeholders
- Risk of overfitting is controlled using max depth

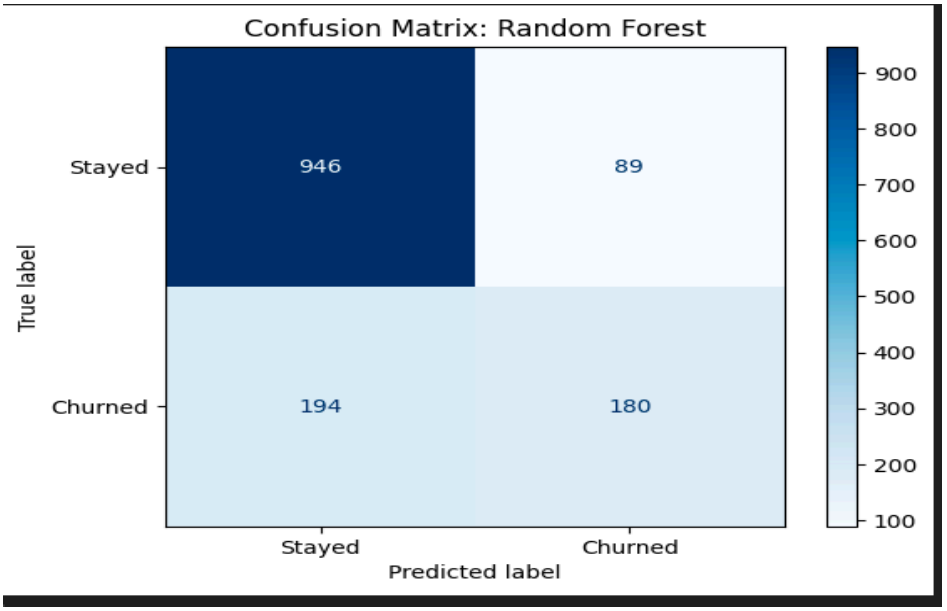
Results:

- **Accuracy:** 79.56%
- **Churn Recall:** 55%

- **Balanced performance**

Good interpretability, decent churn detection

Random Forest



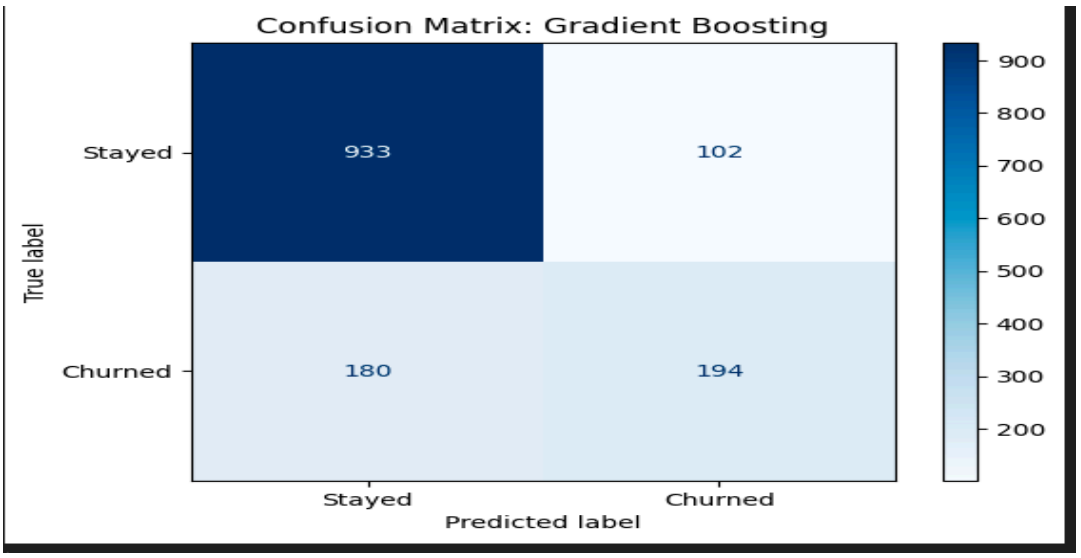
- Combines many decision trees
- Strong at predicting customers who will **stay**
- Slightly weaker in catching churners

Results:

- **Accuracy:** 79.91%
- **Churn Recall:** 48%
- **High precision for non-churn**

Reliable but conservative in predicting churn

Gradient Boosting



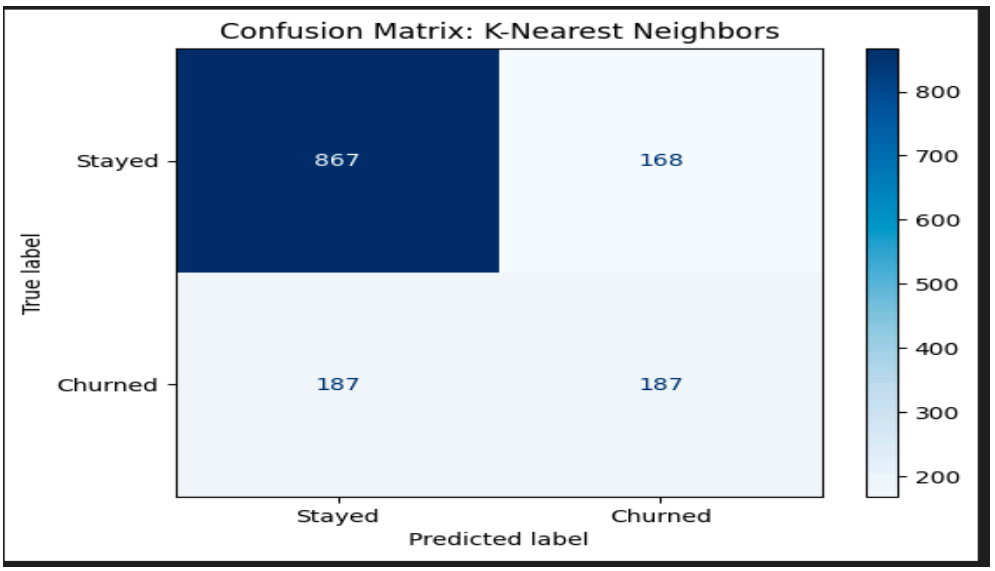
- Focuses on correcting mistakes
- Handles complex feature interactions
- Strong overall balance

Results:

- **Accuracy:** 79.99%
- **Churn Recall:** 52%
- **Good precision-recall balance**

Strong competitor to logistic regression

K-Nearest Neighbors (Worst performer)



- Relies heavily on distance metrics
- Sensitive to feature scaling and high dimensionality
- Struggles with many encoded categorical variables

Results:

- **Accuracy:** 74.80%
- **Churn Recall:** 50%
- **Lowest F1-score**

Not suitable for this dataset

Performance Comparison & Evaluation

Accuracy Comparison

Model	Accuracy

Logistic Regression	**80.48%**
Gradient Boosting	79.99%
Random Forest	79.91%
Decision Tree	79.56%
KNN	74.80%

Churn Recall Comparison (MOST IMPORTANT)

Model	**Churn Recall**
Logistic Regression	**56%**
Decision Tree	55%
Gradient Boosting	52%
KNN	50%
Random Forest	48%

Recall is critical because **missing churners costs the company money**

Final Evaluation

Best Model for This Project: Logistic Regression

Why?

- Highest accuracy
- Best churn recall
- Best F1-score for churn
- Stable, interpretable, business-friendly

Second Best: Gradient Boosting

- Strong overall performance
- Good at learning complex patterns

Worst Choice: KNN

- Poor accuracy
- Weak churn detection
- Not scalable for telecom data

Conclusion

Telecom churn prediction is a binary classification problem aimed at identifying whether a customer will stay with the service or churn. To address this problem in a systematic and business-focused manner, the CRISP-DM methodology was adopted, ensuring a structured workflow from data understanding to model evaluation. Multiple classification algorithms were implemented and compared to assess their accuracy, robustness, and ability to predict customer churn effectively. Python and its rich ecosystem of data science libraries provided a powerful and scalable machine learning solution for data preprocessing, modeling, and evaluation. This approach enables telecom companies to proactively identify customers at risk of leaving and take timely retention actions. Among the five models evaluated, Logistic Regression achieved the best overall performance, delivering the highest accuracy and strong churn recall. Due to its effectiveness in identifying churn-prone customers while maintaining model interpretability and stability, Logistic Regression was selected as the final model for customer churn prediction.

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